

Introduction to Mathematical Reasoning

Week-1

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1 Session Theme

Logical Fundamentals and Proof Warm-up This session introduces the language of mathematical reasoning — propositions, logical connectives, truth tables, and basic proof strategies. It sets the foundation for formal thinking and rigorous argumentation.

2 Learning Objectives

By the end of this session, students will be able to:

- Identify and classify propositions.
- Understand and apply logical connectives.
- Construct and interpret truth tables.
- Recognize valid and invalid logical forms.
- Apply basic proof techniques: direct, contrapositive, contradiction.

3 Propositions and Logical Connectives

3.1 Definition: Proposition

A **proposition** is a declarative sentence that is either **true** or **false**, but not both.

3.2 Examples

- “2 is an even number.” (True)
- “The square root of 2 is rational.” (False)
- “What time is it?” (Not a proposition)

3.3 Logical Connectives

Let P and Q be propositions. We define:

- **Negation:** $\neg P$ (“not P ”)
- **Conjunction:** $P \wedge Q$ (“ P and Q ”)
- **Disjunction:** $P \vee Q$ (“ P or Q ”)
- **Implication:** $P \rightarrow Q$ (“if P , then Q ”)
- **Biconditional:** $P \leftrightarrow Q$ (“ P if and only if Q ”)

4 Truth Tables

4.1 Basic Truth Table for P and Q

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \rightarrow Q$	$P \leftrightarrow Q$
T	T	F	T	T	T	T
T	F	F	F	T	F	F
F	T	T	F	T	T	F
F	F	T	F	F	T	T

4.2 Observation

The implication $P \rightarrow Q$ is only false when P is true and Q is false. The biconditional $P \leftrightarrow Q$ is true when P and Q have the same truth value.

5 Logical Equivalence

5.1 Definition

Two propositions P and Q are **logically equivalent** if $P \leftrightarrow Q$ is a tautology (always true).

5.2 Examples

- $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$ (De Morgan’s Law)
- $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$
- $P \rightarrow Q \equiv \neg P \vee Q$

6 Basic Proof Techniques

6.1 Direct Proof

To prove $P \rightarrow Q$, assume P is true and show that Q follows logically.

6.2 Proof by Contrapositive

Instead of proving $P \rightarrow Q$, prove $\neg Q \rightarrow \neg P$.

6.3 Proof by Contradiction

Assume the negation of the conclusion and derive a contradiction.

6.4 Mathematical Induction (Preview)

Used to prove statements about integers. Will be covered in Week 2.

7 Proof Warm-up

7.1 Example 1: Direct Proof

Claim: If n is even, then n^2 is even.

Proof: Assume n is even. Then $n = 2k$ for some integer k . Then $n^2 = (2k)^2 = 4k^2 = 2(2k^2)$, which is even. $\square$

7.2 Example 2: Contrapositive

Claim: If n^2 is odd, then n is odd.

Proof: We prove the contrapositive: If n is even, then n^2 is even. This is true by Example 1. Therefore, the original statement holds. $\square$

7.3 Example 3: Contradiction

Claim: $\sqrt{2}$ is irrational.

Proof Sketch: Assume $\sqrt{2}$ is rational: $\sqrt{2} = \frac{a}{b}$ in lowest terms. Then $2 = \frac{a^2}{b^2} \Rightarrow a^2 = 2b^2$. So a^2 is even $\Rightarrow a$ is even $\Rightarrow a = 2k$. Substitute: $(2k)^2 = 4k^2 = 2b^2 \Rightarrow b^2 = 2k^2 \Rightarrow b$ is even. Contradiction: both a and b are even $\Rightarrow$ not in lowest terms. $\square$

8 Philosophical Reflection

What makes a mathematical statement “true”? Is truth in mathematics discovered or constructed? How does formal logic differ from everyday reasoning?

9 Homework

1. Construct truth tables for the following:

- $(P \vee Q) \rightarrow R$
- $\neg(P \wedge \neg Q)$

2. Prove: If n is divisible by 4, then n^2 is divisible by 16.

3. Write a short paragraph explaining the difference between “if” and “if and only if”.

4. Challenge: Prove that the sum of two rational numbers is rational.